CONCEPT NOTE | INTERNSHIP CAPSTONE

KhetSathi: AI-assisted crop risk and irrigation guidance

A human-supervised decision-support concept for smallholder tomato growers

Project summary

KhetSathi is a proposed mobile-first assistant that helps tomato growers and extension workers combine crop observations with local weather and field information. A machine-learning model will estimate disease risk for a narrowly defined pilot scope. An AI layer will explain the result in plain language using locally reviewed guidance. An agentic workflow will collect missing inputs, call approved tools, prepare a next-step plan, and schedule a follow-up. Farmers and advisors remain responsible for consequential treatment decisions.

Project at a glance

Primary users	Smallholder tomato growers; extension workers and agronomists as reviewers.
Core problem	Timely, trustworthy disease and irrigation guidance can be difficult to assemble from field symptoms, weather, and local advice.
AI components	ML disease-risk scoring; generative explanation grounded in approved guidance; tool-using agent workflow with human confirmation.
SDGs	SDG 2, target 2.4; SDG 6, target 6.4; SDG 13, target 13.1. These are intended contribution pathways, not claimed outcomes.

1. Problem and rationale

A grower may notice leaf symptoms without knowing whether the cause is a disease, nutrient stress, or another issue. Weather changes can affect disease pressure and irrigation needs, but observations and advice may be scattered across people and tools. A digital assistant could make the information easier to organize and help users decide when to seek expert review. The project will test this workflow in a small, partner-selected pilot rather than claim that an app alone improves yields or saves water.

The SDG link is specific: SDG 2 target 2.4 calls for resilient agricultural practices; SDG 6 target 6.4 addresses water-use efficiency; SDG 13 target 13.1 concerns resilience to climate-related hazards. KhetSathi will measure only pilot-level indicators relevant to these pathways.

2. Aim and objectives

Aim: Design and evaluate a human-supervised AI decision-support prototype for early tomato disease-risk awareness and irrigation prompts.

• Co-design a limited set of crop observations and approved advice with local agricultural reviewers.

• Build a baseline disease-risk model and document its data, limits, and validation.

• Add a grounded conversational layer and an agent workflow that can collect inputs, invoke tools, and propose follow-ups.

• Evaluate model quality, workflow safety, usability, and feasibility with a small pilot; report results without overstating generalizability.

3. Proposed solution and user journey

The prototype is scoped to tomato growers. Pilot geography, disease classes, language, and partner will be selected with an extension or research partner before data collection. The first release will support disease-risk triage and weather-aware irrigation prompts; it will not diagnose every crop condition.

Step	System behavior	Human role
1. Observe	Ask for crop stage, symptom checklist, recent rain/irrigation, and available weather or soil inputs. Flag missing or inconsistent information.	Farmer enters or confirms field observations.
2. Estimate	Call the disease-risk model and irrigation rules/model; show confidence and the factors used.	Advisor can review uncertain or high-risk cases.
3. Explain	Generate a short, localized explanation grounded in approved extension guidance and identify what the model cannot determine.	Farmer decides whether to follow advice or request help.
4. Follow up	Agent drafts a reminder or referral and records the proposed action. No external message or treatment order is sent without confirmation.	User confirms, edits, or rejects the next step.

4. Technical approach

Machine learning: begin with a transparent baseline classifier for a narrowly agreed disease-risk label, using agronomist-labelled observations and permitted weather/field variables. Compare with a simple rules baseline. Split evaluation by farm or time where feasible to reduce leakage. Report class-wise precision, recall, F1, calibration, and the cost of false negatives/positives; do not use synthetic data as evidence of real-world performance.

AI explanation: a language model may summarize the model result, explain relevant observations, and answer questions only from a curated, versioned guidance set. The system should show source snippets and express uncertainty. Unsupported diagnosis or pesticide recommendations are blocked or escalated.

Agentic AI: a bounded orchestrator checks for missing information, calls weather/model/knowledge tools, drafts a plan, and offers a user-confirmed reminder or referral. Tool permissions are narrow, actions are logged, and the agent cannot autonomously order inputs, send messages, or make high-impact treatment decisions.

5. Data, evaluation, and safeguards

Data plan: co-design the data fields with an agricultural partner. Collect only what is necessary, obtain informed consent, use coded identifiers, restrict access, and define deletion and retention periods. Image collection is optional and should be added only if the partner can support reliable labels and consent. External weather sources and any public datasets must be checked for license, coverage, and suitability before use.

Area	Measures	Interpretation
ML model	Class-wise precision/recall/F1, calibration, false-negative review, subgroup and time-split checks.	Report on held-out pilot data; no performance target until baseline and labels are known.
Agent safety	Correct tool calls; evidence-grounded answers; escalation rate; unauthorized external actions.	Target zero unconfirmed external actions; inspect failures manually.
Usability	Task completion, comprehension, time, user feedback, advisor workload.	Small pilot measures workflow fit, not population-wide adoption.
SDG-related	Irrigation prompts accepted; water-use proxy where feasible; timely expert referral; practice changes.	Use a comparison or baseline if feasible; do not infer water savings or yield impact from prompts alone.

Safeguards

- Display uncertainty and the limits of the model; make it easy to request human review.
- Ground generated text in approved sources; log model and knowledge versions.
- Use consent, data minimization, access controls, and a documented retention plan.
- Test language and accessibility with intended users; provide a non-digital or assisted route.
- Treat pesticide, health, financial, or irreversible recommendations as out of scope for autonomous action.

6. SDG alignment

SDG 2, target 2.4: support the exploration of resilient and sustainable agricultural practices. **SDG 6, target 6.4:** test whether weather-aware prompts help users make more informed water-use decisions. **SDG 13, target 13.1:** improve access to climate-related risk information and escalation pathways. These links describe project intent; contribution would require field evidence.

7. Delivery plan and risks

Indicative 10-week internship plan; adjust to the actual internship calendar and partner access.

Weeks	Work	Output
1-2	Problem interviews, partner selection, scope and consent/data plan.	Validated problem statement and pilot protocol.
3-4	Data schema, approved guidance, baseline and rules benchmark.	Model/data card and evaluation plan.
5-7	Prototype user flow, model tool, grounded AI layer, bounded agent and logging.	Demonstrable MVP with safeguards.
8-9	Advisor review, usability sessions, held-out evaluation and failure analysis.	Results and limitations.
10	Refine, document, present and plan next validation stage.	Final project, Lean Canvas, Concept Note, presentation.

Key risks and responses

Risk	Response
Sparse or biased labels	Start with a small bounded scope; involve an agronomist; disclose coverage gaps; fall back to referral rather than overconfident output.
Connectivity or language barriers	Test low-bandwidth flows, local-language wording, and assisted/non-digital alternatives.
Hallucinated or unsafe advice	Retrieve only from approved guidance, show source and uncertainty, constrain tool calls, and require human confirmation.
No measurable water/yield effect	Treat outcomes as hypotheses; use process measures first and avoid causal claims without a suitable study design.

8. Expected outputs

• A functional, bounded prototype and documented user journey.

• A transparent baseline model evaluation on appropriately labelled pilot data, if data access is secured; otherwise a clearly labelled design prototype.

• A data card, model card, agent action log, privacy/safety notes, and user feedback summary.

• Three internship submissions: Lean Canvas PDF, Concept Note, and PowerPoint presentation.

Assumptions to confirm before implementation: local pilot partner, district/crop practices, supported language, data access, approved agricultural guidance, and internship timeline.

References

United Nations Department of Economic and Social Affairs. Sustainable Development Goal 2: Zero Hunger; target 2.4. https://sdgs.un.org/goals/goal2

United Nations Department of Economic and Social Affairs. Sustainable Development Goal 6: Clean Water and Sanitation; target 6.4. https://sdgs.un.org/goals/goal6

United Nations Department of Economic and Social Affairs. Sustainable Development Goal 13: Climate Action; target 13.1. https://sdgs.un.org/goals/goal13

IBM SkillsBuild, Masterclass 5: Creating Real Life Projects - Project Guidance and Creating a Lean Canvas (user-provided sample).